PHYS551: Advanced Monte Carlo Methods

Homework 2

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Disclosures and References

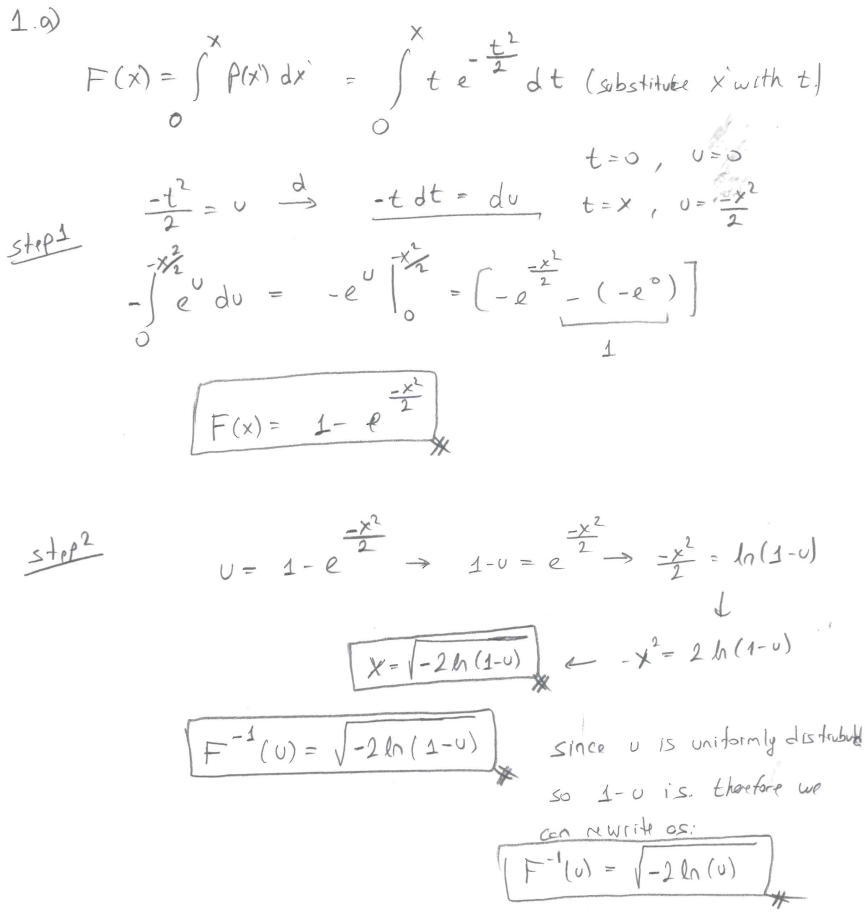
AI Usage Disclosure: In accordance with the course policy, I would like to declare the extent of AI assistance used in this assignment:

- **Analytical Derivations:** All mathematical derivations were performed manually without the use of AI tools.
- **Code Implementation:** The code for all problems was written by me without direct copying from AI-generated sources. However, I consulted for commenting the documentation and checking the code if there were any mistakes.
- **Report Writing:** The explanations, discussions, and formatting of this report were written by me. AI tools were used to check for grammatical errors and to give feedback on the overall structure and style.

Problem 1: The Custom Generator (Inverse Transform)

1. Analytical Derivation

The derivation for the Inverse Transform method is done by hand and is shown in the figure below.



1.a)

$$F(x) = \int_0^x p(x')\,dx' = \int_0^x t\,e^{-\frac{t^2}{2}}\,dt \quad \text{(substitute } x' \text{ with } t\text{)}$$

step1

$$\frac{-t^2}{2} = u \xrightarrow{d} -t\,dt = du \qquad t=0,\ u=0 \qquad t=x,\ u=\frac{-x^2}{2}$$

$$-\int_0^{-x^2/2} e^u\,du = -e^u \Big|_0^{-x^2/2} = \Big[-e^{\frac{-x^2}{2}} \underbrace{-(-e^0)}_{1}\Big]$$

$$\boxed{F(x) = 1 - e^{\frac{-x^2}{2}}}$$

step2

$$u = 1 - e^{\frac{-x^2}{2}} \rightarrow 1-u = e^{\frac{-x^2}{2}} \rightarrow \frac{-x^2}{2} = \ln(1-u)$$

$$\downarrow$$

$$\boxed{x = \sqrt{-2\ln(1-u)}} \leftarrow -x^2 = 2\ln(1-u)$$

$$\boxed{F^{-1}(u) = \sqrt{-2\ln(1-u)}}$$

since u is uniformly distributed so 1-u is. therefore we can rewrite as:

$$\boxed{F^{-1}(u) = \sqrt{-2\ln(u)}}$$

Figure 1: Handwritten derivation for Inverse Transform of Rayleigh Distribution

2. Implementation

Based on the analytical derivation above, the random variable X following a Rayleigh distribution can be generated by applying the transformation $x = \sqrt{-2\ln(U)}$ to a standard uniform random variable $U \sim \text{Uniform}(0, 1)$.

The following implementation defines a generator function and verifies it by comparing the empirical distribution of $10,000$ samples against the theoretical probability density function (PDF).

Python Source Code for Problem 1

```python
"""

# PHYS 551: Advanced Monte Carlo Methods
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"""

import numpy as np
import matplotlib.pyplot as plt


"""Problem 1: The Custom Generator (Inverse Transform)"""

# Part (b): Function to generate N Rayleigh samples
def generate_rayleigh(N):
    """Generates N random samples from a Rayleigh distribution using Inverse Transform."""
    N_sample = np.random.uniform(0, 1, N)
    return np.sqrt(-2 * np.log(N_sample))

# Part (c): Function to compute the empirical CDF and showing theoretical CDF
def plot_cdf(N):

    samples = generate_rayleigh(N) # Generate N Rayleigh samples from the function defined in part (b)
    plt.hist(samples, bins=50, density=True, alpha=0.6, color='b', label='Empirical PDF (Histogram)')

    x_values = np.linspace(0, 6, 500) # Generate x values for plotting the theoretical PDF

    # p(x) = x * exp(-x^2 / 2) for x >= 0
    pdf_theoretical = x_values * np.exp(-x_values**2 / 2)

    plt.plot(x_values, pdf_theoretical, lw=2,color='red', label='Theoretical PDF $p(x)$')

    plt.title('Verification of Rayleigh Distribution Generator')
    plt.xlabel('x')
    plt.ylabel('Probability Density')
    plt.legend()
    plt.grid(axis='y', alpha=0.3)
    plt.show()

def problem_1_b():
    samples = generate_rayleigh(10000)  # Generate 10,000 samples
    print("First 10 samples from Rayleigh distribution:", samples[:10])

def problem_1_c():
    N = 10000
    plot_cdf(N)


def run_problem_1():
    print("Problem 1(b): Generating Rayleigh samples")
    problem_1_b()
    print("\nProblem 1(c): Plotting PDF")
    problem_1_c()

def main():
    run_problem_1()

main()
```

3. Verification and Results

To verify the accuracy of the generator, $N = 10,000$ samples were generated and plotted as a normalized histogram. The theoretical PDF, $p(x) = xe^{-x^2/2}$, is overlaid as a solid red line for comparison.

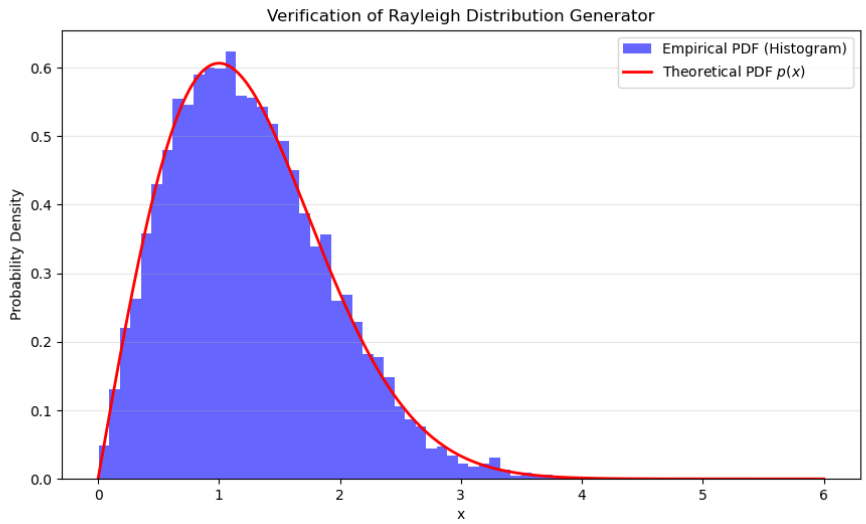

Figure 2: Normalized histogram of 10,000 Rayleigh samples vs. Theoretical PDF.

Observations: As shown in Figure 2, the empirical distribution of the generated samples (blue bars) closely matches the theoretical Rayleigh density function (red curve). The peak of the distribution occurs near $x = 1$, which is consistent with the mode of the Rayleigh distribution.

Problem 2: Working Smarter (Variance Reduction)

In this problem, we estimate the integral $I = \int_0^1 e^{x^2} dx \approx 1.46265$ using three different Monte Carlo approaches with $N = 10,000$ samples.

1. Control Variates: Analytical Derivation

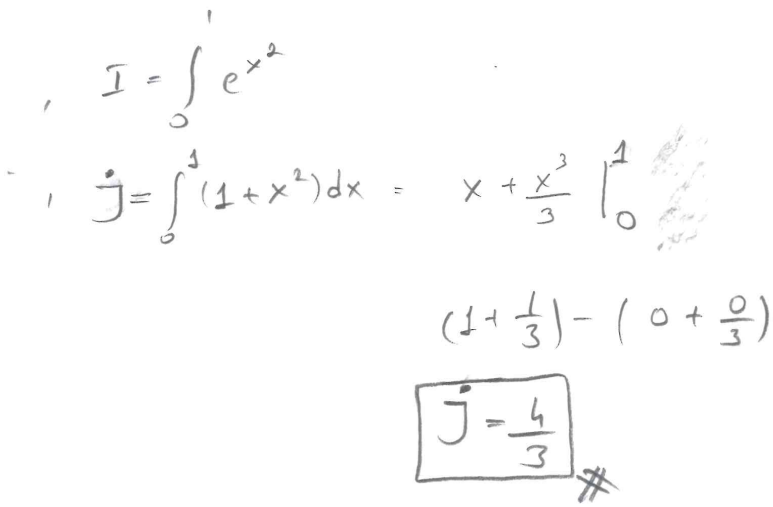

Figure 3: Handwritten derivation for Control Variates Method

2. Implementation

The three methods (Naive, Antithetic, and Control Variates) were implemented in Python. The Antithetic method utilizes pairs $(U, 1 - U)$ to introduce negative correlation, while the Control Variates method uses the derivation above to reduce variance.

Python Source Code for Problem 2

```python
"""

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"""

import numpy as np
import matplotlib.pyplot as plt


"""Problem 2: Working Smarter (Variance Reduction)"""

# Part (a): Naive Monte Carlo method
def naive_method(N):
    samples = np.random.uniform(0, 1, N) # N samples from uniform distribution in [0, 1]
    result = np.exp(samples**2) # Compute f(x) = exp(x^2) for each sample

```

```python
    integral_estimate = np.mean(result)
    variance = np.var(result, ddof=1)
    sem = np.sqrt(variance / N)

    return integral_estimate, variance, sem

# Part (b): Antithetic Variates method
def antithetic_method(N):
    sample_1 = np.random.uniform(0, 1, N//2) # N/2 samples from uniform distribution in [0, 1]
    sample_2 = 1 - sample_1 # Antithetic samples

    result_1 = np.exp(sample_1**2) # Compute f(x) for the first set of samples
    result_2 = np.exp(sample_2**2) # Compute f(x) for the antithetic samples

    pair_mean = (result_1 + result_2) / 2 # Average the results of the pairs

    intgral_estimate = np.mean(pair_mean)
    variance = np.var(pair_mean, ddof=1)
    sem = np.sqrt(variance / (N//2))

    return intgral_estimate, variance, sem

# Part (c): Control Variates method
def control_variates_method(N):

    samples  = np.random.uniform(0, 1, N)

    X = np.exp(samples ** 2) # f(X) = e^(X^2)
    Y = samples ** 2 + 1 # g(X) = X^2 + 1, J = E[g(X)] = 4/3

    J = 4 / 3

    c_estimated =  np.cov(X, Y)[0, 1] / np.var(Y, ddof=1) # Estimate the optimal control variate coefficient c

    L = X - c_estimated * (Y - J) # Adjusted estimator using control variates

    mean = np.mean(L)
    variance = np.var(L, ddof=1)
    sem = np.sqrt(variance / N)

    return mean, variance, sem

def run_problem_2():
    N = 10000

    # Run all three methods and collect their estimates, variances, and SEMs
    naive_est, naive_var, naive_sem = naive_method(N)
    anti_est, anti_var, anti_sem = antithetic_method(N)
    control_est, control_var, control_sem = control_variates_method(N)

    # Calculate Variance Reduction Factors (VRF) for both methods compared to the naive method
    vrf_anti = naive_var / anti_var
    vrf_control = naive_var / control_var

    # Print the results in a formatted table
    print("\nProblem 2 Results (N = 10,000)")
    print("-" * 65)
    print(f"{'Method':<22} {'Estimate':<12} {'Variance':<14} {'SEM':<14}")
    print("-" * 65)

    print(f"{'Naive Monte Carlo':<22} {naive_est:<12.6f} {naive_var:<14.6e} {naive_sem:<14.6e}")
    print(f"{'Antithetic Variates':<22} {anti_est:<12.6f} {anti_var:<14.6e} {anti_sem:<14.6e} ")
    print(f"{'Control Variates':<22} {control_est:<12.6f} {control_var:<14.6e} {control_sem:<14.6e}")

    print("-" * 65)
    print(f"VRF (Antithetic vs Naive): {vrf_anti:.3f}")
    print(f"VRF (Control vs Naive):    {vrf_control:.3f}")
    print("-" * 65)

def main():
    run_problem_2()

main()
```

3. Verification and Results

Problem 2 Results

```
Problem 2 Results (N = 10,000)
----------------------------------------------------------------
Method Estimate Variance SEM
----------------------------------------------------------------
Naive Monte Carlo 1.460129 2.256655e-01 4.750426e-03
Antithetic Variates 1.464030 2.779314e-02 2.357674e-03
Control Variates 1.461401 3.738740e-03 6.114523e-04
----------------------------------------------------------------
VRF (Antithetic vs Naive):  8.119
VRF (Control vs Naive):  60.359
----------------------------------------------------------------
```

The results of the simulation are tabulated below. For all methods, the sample size was kept constant at $N = 10,000$ to provide a fair comparison of efficiency.

Method	**Estimate** $\hat{I}$	**Variance** (σ^2)	**SEM**	**VRF**
Naive Monte Carlo	1.4601	225.67×10^{-3}	4.75×10^{-3}	1.00
Antithetic Variates	1.4640	27.79×10^{-3}	2.36×10^{-3}	8.12
Control Variates	1.4614	3.74×10^{-3}	0.61×10^{-3}	60.36

Table 1: Performance comparison of variance reduction techniques for $N = 10,000$.

Observations: The data clearly demonstrates that **Control Variates** is the most effective method for this integral. While the Antithetic method provided a significant gain by reducing variance by a factor of 8.12, the Control Variates method achieved a Variance Reduction Factor (VRF) of 60.36. This implies that the Control Variates estimator is roughly 60 times more computationally efficient than the Naive approach.

Problem 3: The Curse of Dimensionality

1. Simulation Implementation

The estimation of a d-dimensional hypersphere volume is performed using rejection sampling (the "dart-throwing" method). We generate N points in a d-dimensional hypercube of side length 2 (range $[-1, 1]$) and count how many fall within the unit distance from the origin.

Python Source Code for Problem 3

```python
"""

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"""
import numpy as np

"""Problem 3: The Curse of Dimensionality"""

def estimate_volume(d, n_samples=1_000_000):
    # Generate N points in d-dimensions within [-1, 1]
    points = np.random.uniform(-1, 1, (n_samples, d))

    # Calculate squared Euclidean distance for each point
    distances_squared = np.sum(points**2, axis=1)

    # Count points satisfying the hypersphere condition: sum(x_i^2) <= 1
    inside_count = np.sum(distances_squared <= 1)

    # Volume of the bounding hypercube is 2^d
    v_cube = 2**d

    # Scale the hit-ratio by the total volume of the cube
    estimated_volume = (inside_count / n_samples) * v_cube

    return estimated_volume

def run_problem_3():
    dimensions = [2, 5, 10]
    results = {d: estimate_volume(d) for d in dimensions}

    print(f"Problem 3 Results (N = 1,000,000 samples):")
    print("-" * 45)
    print(f"{'Dimension (d)':<15} {'Estimated Volume':<20}")
    print("-" * 45)
    for d, vol in results.items():
        print(f"{d:<15} {vol:<20.5f}")

def main():
    run_problem_3()

if __name__ == "__main__":
    main()
```

2. Results and Discussion

The simulation was conducted for $d \in \{2, 5, 10\}$ using $N = 10^6$ samples. The results are summarized in Table 2.

Table 2: Hypersphere Volume Estimates ($N = 1,000,000$)

Dimension (d)	Estimated Volume
2	3.14150
5	5.26384
10	2.54874

Analysis of Findings

- **Accuracy:** The 10D estimate (≈ 2.549) is highly consistent with the analytical value $V_{10} = \frac{\pi^5}{120} \approx 2.550$.
- **Curse of Dimensionality:** At $d = 10$, the ratio of the hypersphere volume to the bounding hypercube ($2^{10} = 1024$) is a mere 0.249%. This illustrates that in high-dimensional spaces, volume concentrates in the "corners" of the hypercube, leaving the inscribed sphere nearly negligible.
- **Computational Efficiency:** From a Monte Carlo perspective, over 99.7% of samples are rejected at $d = 10$. Because the "hit rate" decays exponentially with d, the sample size N must grow exponentially to maintain precision, rendering simple rejection sampling impractical for high-dimensional integration.

Output

Sample Terminal Output from Running the Code

```
Problem 1(b):  Generating Rayleigh samples
First 10 samples from Rayleigh distribution:  [1.10402963 1.16078819 1.66809063 1.72289481 1.95939379 2.17591001
1.10808643 0.57249355 0.46745059 1.33945967]
Problem 1(c):  Plotting PDF
Problem 2 Results (N = 10,000)
----------------------------------------------------------------
Method Estimate Variance SEM
----------------------------------------------------------------
Naive Monte Carlo 1.451307 2.182678e-01 4.671914e-03
Antithetic Variates 1.462003 2.772500e-02 2.354782e-03
Control Variates 1.462234 3.797647e-03 6.162505e-04
----------------------------------------------------------------
VRF (Antithetic vs Naive):  7.873
VRF (Control vs Naive):  57.474
----------------------------------------------------------------
Problem 3 Results (N = 1,000,000 samples):
-------------------------------------------
Dimension (d) Estimated Volume
-------------------------------------------
2 3.14163
5 5.26656
10 2.47398
```